INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 7: References

by

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7 References

Almasoud, A. S., Hussain, F. K., & Hussain, O. K. (2020). Smart contracts for blockchain-based reputation systems: A systematic literature review. *Journal of Network and Computer Applications*, *170*, Article 102814. https://doi.org/10.1016/j.jnca.2020.102814

Angeris, G., & Chitra, T. (2020). Improved price oracles: Constant function market makers. *Proceedings of the 2nd ACM Conference on Advances in Financial Technologies*, 80–91. https://doi.org/10.1145/3419614.3423251

Antonopoulos, A. M., & Wood, G. (2018). *Mastering Ethereum: Building smart contracts and DApps*. O'Reilly Media.

Ballet, G., Buterin, V., & Feist, D. (2023). *EIP-6780: SELFDESTRUCT only in same transaction* (Ethereum Improvement Proposal No. 6780). Ethereum Improvement Proposals. https://eips.ethereum.org/EIPS/eip-6780

Barabási, A.-L., & Albert, R. (1999). Emergence of scaling in random networks. *Science*, *286*(5439), 509–512. https://doi.org/10.1126/science.286.5439.509

Bartoletti, M., Chiang, J. H.-y., & Lluch Lafuente, A. (2021). SoK: Lending pools in decentralized finance. In M. Bernhard, A. Bracciali, L. Gudgeon, T. Haines, A. Klages-Mundt, S. Matsuo, D. Perez, M. Sala, & S. Werner (Eds.), *Financial cryptography and data security: FC 2021 international workshops* (pp. 553–578). Springer. https://doi.org/10.1007/978-3-662-63958-0_40

Bellini, E., Iraqi, Y., & Damiani, E. (2020). Blockchain-based distributed trust and reputation management systems: A survey. *IEEE Access*, *8*, 21127–21151. https://doi.org/10.1109/ACCESS.2020.2969820

Bordeianu, A. A., & Popescu, D. E. (2026). Threat landscape in decentralized systems: Sybil attacks, related vulnerabilities, and blockchain security evolution (2015–2025). *Applied Sciences*, *16*(14), Article 6929. https://doi.org/10.3390/app16146929

Buterin, V. (2014). *Ethereum: A next-generation smart contract and decentralized application platform* [White paper]. Ethereum Foundation. https://ethereum.org/en/whitepaper/

Buterin, V., Wilson, S., Dietrichs, A., & lightclient. (2024). *EIP-7702: Set code for EOAs* (Ethereum Improvement Proposal No. 7702). Ethereum Improvement Proposals. https://eips.ethereum.org/EIPS/eip-7702

Chainlink. (n.d.). *Selecting quality data feeds* [Chainlink documentation]. Retrieved October 10, 2026, from https://docs.chain.link/data-feeds/selecting-data-feeds

Cheng, A., & Friedman, E. (2005). Sybilproof reputation mechanisms. *Proceedings of the 2005 ACM SIGCOMM Workshop on Economics of Peer-to-Peer Systems*, 128–132. https://doi.org/10.1145/1080192.1080202

Cong, L. W., Li, X., Tang, K., & Yang, Y. (2023). Crypto wash trading. *Management Science*, *69*(11), 6427–6454. https://doi.org/10.1287/mnsc.2021.02709

Cronbach, L. J., & Meehl, P. E. (1955). Construct validity in psychological tests. *Psychological Bulletin*, *52*(4), 281–302. https://doi.org/10.1037/h0040957

Daian, P., Goldfeder, S., Kell, T., Li, Y., Zhao, X., Bentov, I., Breidenbach, L., & Juels, A. (2020). Flash Boys 2.0: Frontrunning in decentralized exchanges, miner extractable value, and consensus instability. *2020 IEEE Symposium on Security and Privacy (SP)*, 910–927. https://doi.org/10.1109/SP40000.2020.00040

De Domenico, M., Solé-Ribalta, A., Omodei, E., Gómez, S., & Arenas, A. (2015). Ranking in interconnected multilayer networks reveals versatile nodes. *Nature Communications*, *6*, Article 6868. https://doi.org/10.1038/ncomms7868

DefiLlama. (n.d.). *DefiLlama API documentation* [Coins endpoints]. Retrieved October 10, 2026, from https://api-docs.defillama.com/

DiCiccio, T. J., & Efron, B. (1996). Bootstrap confidence intervals. *Statistical Science*, *11*(3), 189–228. https://doi.org/10.1214/ss/1032280214

Do, H.-D., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. *International Journal of Software Innovation*, *11*(1), 1–13. https://doi.org/10.4018/IJSI.315737

Do, T., Do, D., & Nguyen, L. (2023). On-chain reputation ranking by Adaptive Weighted PageRank. *2023 RIVF International Conference on Computing and Communication Technologies (RIVF)*, 318–323. https://doi.org/10.1109/RIVF60135.2023.10471809

Douceur, J. R. (2002). The Sybil attack. In P. Druschel, F. Kaashoek, & A. Rowstron (Eds.), *Peer-to-peer systems* (pp. 251–260). Springer. https://doi.org/10.1007/3-540-45748-8_24

Efron, B. (1979). Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, *7*(1), 1–26. https://doi.org/10.1214/aos/1176344552

Fortunato, S., Boguñá, M., Flammini, A., & Menczer, F. (2008). Approximating PageRank from in-degree. In W. Aiello, A. Broder, J. Janssen, & E. Milios (Eds.), *Algorithms and models for the web-graph: WAW 2006* (pp. 59–71). Springer. https://doi.org/10.1007/978-3-540-78808-9_6

Fritsch, R., Müller, M., & Wattenhofer, R. (2024). Analyzing voting power in decentralized governance: Who controls DAOs? *Blockchain: Research and Applications*, *5*(3), Article 100208.

Gervais, A., Karame, G. O., Wüst, K., Glykantzis, V., Ritzdorf, H., & Capkun, S. (2016). On the security and performance of proof of work blockchains. *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security*, 3–16. https://doi.org/10.1145/2976749.2978341

GMX. (n.d.). *GMX docs*. Retrieved October 9, 2026, from https://docs.gmx.io/

Gudgeon, L., Moreno-Sanchez, P., Roos, S., McCorry, P., & Gervais, A. (2020). SoK: Layer-two blockchain protocols. In J. Bonneau & N. Heninger (Eds.), *Financial cryptography and data security: FC 2020* (pp. 201–226). Springer. https://doi.org/10.1007/978-3-030-51280-4_12

Gudgeon, L., Perez, D., Harz, D., Livshits, B., & Gervais, A. (2020). The decentralized financial crisis. *2020 Crypto Valley Conference on Blockchain Technology (CVCBT)*, 1–15. https://doi.org/10.1109/CVCBT50464.2020.00005

Gyöngyi, Z., & Garcia-Molina, H. (2005). Link spam alliances. *Proceedings of the 31st International Conference on Very Large Data Bases*, 517–528. https://www.vldb.org/conf/2005/papers/p517-gyongyi.pdf

Gyöngyi, Z., Garcia-Molina, H., & Pedersen, J. (2004). Combating web spam with TrustRank. *Proceedings of the Thirtieth International Conference on Very Large Data Bases*, 576–587. https://doi.org/10.1016/B978-012088469-8.50052-8

Halu, A., Mondragón, R. J., Panzarasa, P., & Bianconi, G. (2013). Multiplex PageRank. *PLoS ONE*, *8*(10), Article e78293. https://doi.org/10.1371/journal.pone.0078293

Harvey, C. R., Ramachandran, A., & Santoro, J. (2021). *DeFi and the future of finance*. Wiley.

Hassija, V., Bansal, G., Chamola, V., Kumar, N., & Guizani, M. (2020). Secure lending: Blockchain and prospect theory-based decentralized credit scoring model. *IEEE Transactions on Network Science and Engineering*, *7*(4), 2566–2575. https://doi.org/10.1109/TNSE.2020.2982488

Haveliwala, T. H. (2003). Topic-sensitive PageRank: A context-sensitive ranking algorithm for web search. *IEEE Transactions on Knowledge and Data Engineering*, *15*(4), 784–796. https://doi.org/10.1109/TKDE.2003.1208999

He, B., Chen, Y., Chen, Z., Hu, X., Hu, Y., Wu, L., Chang, R., Wang, H., & Zhou, Y. (2023). TxPhishScope: Towards detecting and understanding transaction-based phishing on Ethereum. *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security*, 120–134. https://doi.org/10.1145/3576915.3623210

Human Passport. (n.d.). *Human Passport developer platform*. Retrieved October 9, 2026, from https://docs.passport.human.tech/

Jeh, G., & Widom, J. (2003). Scaling personalized web search. *Proceedings of the 12th International Conference on World Wide Web*, 271–279. https://doi.org/10.1145/775152.775191

Kalodner, H., Goldfeder, S., Chen, X., Weinberg, S. M., & Felten, E. W. (2018). Arbitrum: Scalable, private smart contracts. *Proceedings of the 27th USENIX Security Symposium*, 1353–1370. https://www.usenix.org/conference/usenixsecurity18/presentation/kalodner

Kamvar, S. D., Schlosser, M. T., & Garcia-Molina, H. (2003). The EigenTrust algorithm for reputation management in P2P networks. *Proceedings of the 12th International Conference on World Wide Web*, 640–651. https://doi.org/10.1145/775152.775242

Kendall, M. G. (1938). A new measure of rank correlation. *Biometrika*, *30*(1/2), 81–93. https://doi.org/10.2307/2332226

Kitzler, S., Victor, F., Saggese, P., & Haslhofer, B. (2023). Disentangling decentralized finance (DeFi) compositions. *ACM Transactions on the Web*, *17*(2), 1–26. https://doi.org/10.1145/3532857

Klages-Mundt, A., Harz, D., Gudgeon, L., Liu, J.-Y., & Minca, A. (2020). Stablecoins 2.0: Economic foundations and risk-based models. *Proceedings of the 2nd ACM Conference on Advances in Financial Technologies*, 59–79. https://doi.org/10.1145/3419614.3423261

Kleinberg, J. M. (1999). Authoritative sources in a hyperlinked environment. *Journal of the ACM*, *46*(5), 604–632. https://doi.org/10.1145/324133.324140

Kotb, M. O. (2023). Credit scoring using machine learning algorithms and blockchain technology. *2023 Intelligent Methods, Systems, and Applications (IMSA)*, 381–386. https://doi.org/10.1109/IMSA58542.2023.10217411

Langville, A. N., & Meyer, C. D. (2006). *Google's PageRank and beyond: The science of search engine rankings*. Princeton University Press. https://doi.org/10.1515/9781400830329

Leskovec, J., Rajaraman, A., & Ullman, J. D. (2020). *Mining of massive datasets* (3rd ed.). Cambridge University Press. https://doi.org/10.1017/9781108684163

Li, W., Bu, J., Li, X., & Chen, X. (2022). Security analysis of DeFi: Vulnerabilities, attacks and advances. *2022 IEEE International Conference on Blockchain (Blockchain)*, 488–493. https://doi.org/10.1109/Blockchain55522.2022.00075

Liben-Nowell, D., & Kleinberg, J. (2007). The link-prediction problem for social networks. *Journal of the American Society for Information Science and Technology*, *58*(7), 1019–1031. https://doi.org/10.1002/asi.20591

Lin, D., Wu, J., Fu, Q., Zheng, Z., & Chen, T. (2025). RiskProp: Account risk rating on Ethereum via de-anonymous score and network propagation. *IEEE Transactions on Dependable and Secure Computing*, *22*(3), 2132–2145. https://doi.org/10.1109/TDSC.2024.3475825

Lin, D., Wu, J., Huang, T., Lin, K., & Zheng, Z. (2024). Who is who on Ethereum? Account labeling using heterophilic graph convolutional network. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, *54*(3), 1541–1553. https://doi.org/10.1109/TSMC.2023.3329520

Lin, D., Wu, J., Xuan, Q., & Tse, C. K. (2022). Ethereum transaction tracking: Inferring evolution of transaction networks via link prediction. *Physica A: Statistical Mechanics and its Applications*, *600*, Article 127504. https://doi.org/10.1016/j.physa.2022.127504

Lin, Z., Chen, J., Wu, J., Zhang, W., Wang, Y., & Zheng, Z. (2024). CRPWarner: Warning the risk of contract-related rug pull in DeFi smart contracts. *IEEE Transactions on Software Engineering*, *50*(6), 1534–1547. https://doi.org/10.1109/TSE.2024.3392451

Liu, X. F., Jiang, X.-J., Liu, S.-H., & Tse, C. K. (2021). Knowledge discovery in cryptocurrency transactions: A survey. *IEEE Access*, *9*, 37229–37254. https://doi.org/10.1109/ACCESS.2021.3062652

Lundfall, M. (2020). *ERC-2612: Permit extension for EIP-20 signed approvals* (Ethereum Improvement Proposal No. 2612). Ethereum Improvement Proposals. https://eips.ethereum.org/EIPS/eip-2612

Luo, J., Kang, H., Zheng, S., & Liu, X. (2025). Toward resilient airdrop mechanisms: Empirical measurement of hunter profits and airdrop game theory modeling. *2025 IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*, 1–5. https://doi.org/10.1109/ICBC64466.2025.11114597

Makarov, I., & Schoar, A. (2020). Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics*, *135*(2), 293–319. https://doi.org/10.1016/j.jfineco.2019.07.001

Messick, S. (1995). Validity of psychological assessment: Validation of inferences from persons' responses and performances as scientific inquiry into score meaning. *American Psychologist*, *50*(9), 741–749. https://doi.org/10.1037/0003-066X.50.9.741

Nakamoto, S. (2008). *Bitcoin: A peer-to-peer electronic cash system* [White paper]. https://bitcoin.org/bitcoin.pdf

Narayanan, A., Bonneau, J., Felten, E., Miller, A., & Goldfeder, S. (2016). *Bitcoin and cryptocurrency technologies: A comprehensive introduction*. Princeton University Press.

Newman, M. E. J. (2001). Clustering and preferential attachment in growing networks. *Physical Review E*, *64*(2), Article 025102. https://doi.org/10.1103/PhysRevE.64.025102

Nosek, B. A., Ebersole, C. R., DeHaven, A. C., & Mellor, D. T. (2018). The preregistration revolution. *Proceedings of the National Academy of Sciences*, *115*(11), 2600–2606. https://doi.org/10.1073/pnas.1708274114

Packin, N. G., & Lev-Aretz, Y. (2024). Decentralized credit scoring: Black box 3.0. *American Business Law Journal*, *61*(2), 91–111. https://doi.org/10.1111/ablj.12240

Page, L., Brin, S., Motwani, R., & Winograd, T. (1999). *The PageRank citation ranking: Bringing order to the web* (Technical Report No. SIDL-WP-1999-0120). Stanford InfoLab. http://ilpubs.stanford.edu:8090/422/

Palaiokrassas, G., Scherrers, S., Makri, E., & Tassiulas, L. (2024). Machine learning in DeFi: Credit risk assessment and liquidation prediction. *2024 IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*, 650–654. https://doi.org/10.1109/ICBC59979.2024.10634435

Perez, D., Werner, S. M., Xu, J., & Livshits, B. (2021). Liquidations: DeFi on a knife-edge. In N. Borisov & C. Diaz (Eds.), *Financial cryptography and data security: FC 2021, Part II* (pp. 457–476). Springer. https://doi.org/10.1007/978-3-662-64331-0_24

Qin, K., Zhou, L., & Gervais, A. (2022). Quantifying blockchain extractable value: How dark is the forest? *2022 IEEE Symposium on Security and Privacy (SP)*, 198–214. https://doi.org/10.1109/SP46214.2022.9833734

Qin, K., Zhou, L., Livshits, B., & Gervais, A. (2021). Attacking the DeFi ecosystem with flash loans for fun and profit. In N. Borisov & C. Diaz (Eds.), *Financial cryptography and data security: FC 2021, Part I* (pp. 3–32). Springer. https://doi.org/10.1007/978-3-662-64322-8_1

Resnick, P., & Zeckhauser, R. (2002). Trust among strangers in internet transactions: Empirical analysis of eBay's reputation system. In M. R. Baye (Ed.), *The economics of the internet and e-commerce* (pp. 127–157). Emerald. https://doi.org/10.1016/S0278-0984(02)11030-3

Schär, F. (2021). Decentralized finance: On blockchain- and smart contract-based financial markets. *Federal Reserve Bank of St. Louis Review*, *103*(2), 153–174. https://doi.org/10.20955/r.103.153-74

Siddarth, D., Ivliev, S., Siri, S., & Berman, P. (2020). Who watches the watchmen? A review of subjective approaches for Sybil-resistance in proof of personhood protocols. *Frontiers in Blockchain*, *3*, Article 590171. https://doi.org/10.3389/fbloc.2020.590171

Spearman, C. (1904). The proof and measurement of association between two things. *The American Journal of Psychology*, *15*(1), 72–101. https://doi.org/10.2307/1412159

Szabo, N. (1997). Formalizing and securing relationships on public networks. *First Monday*, *2*(9). https://doi.org/10.5210/fm.v2i9.548

Thibault, L. T., Sarry, T., & Hafid, A. S. (2022). Blockchain scaling using rollups: A comprehensive survey. *IEEE Access*, *10*, 93039–93054. https://doi.org/10.1109/ACCESS.2022.3200051

Victor, F. (2020). Address clustering heuristics for Ethereum. In J. Bonneau & N. Heninger (Eds.), *Financial cryptography and data security: FC 2020* (pp. 617–633). Springer. https://doi.org/10.1007/978-3-030-51280-4_33

Victor, F., & Weintraud, A. M. (2021). Detecting and quantifying wash trading on decentralized cryptocurrency exchanges. *Proceedings of the Web Conference 2021*, 23–32. https://doi.org/10.1145/3442381.3449824

Vogelsteller, F., & Buterin, V. (2015). *ERC-20: Token standard* (Ethereum Improvement Proposal No. 20). Ethereum Improvement Proposals. https://eips.ethereum.org/EIPS/eip-20

Wang, D., Feng, H., Wu, S., Zhou, Y., Wu, L., & Yuan, X. (2022). Penny wise and pound foolish: Quantifying the risk of unlimited approval of ERC20 tokens on Ethereum. *Proceedings of the 25th International Symposium on Research in Attacks, Intrusions and Defenses*, 99–114. https://doi.org/10.1145/3545948.3545963

Werner, S., Perez, D., Gudgeon, L., Klages-Mundt, A., Harz, D., & Knottenbelt, W. (2022). SoK: Decentralized finance (DeFi). *Proceedings of the 4th ACM Conference on Advances in Financial Technologies*, 30–46. https://doi.org/10.1145/3558535.3559780

Wood, G. (2025, February 4). *Ethereum: A secure decentralised generalised transaction ledger* [Ethereum Yellow Paper] (Shanghai version efc5f9a). https : / / ethereum . github . io / yellowpaper/paper.pdf

Wu, J., Yuan, Q., Lin, D., You, W., Chen, W., Chen, C., & Zheng, Z. (2022). Who are the phishers? Phishing scam detection on Ethereum via network embedding. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, *52*(2), 1156–1166. https://doi.org/10.1109/TSMC.2020.3016821

Xia, P., Wang, H., Gao, B., Su, W., Yu, Z., Luo, X., Zhang, C., Xiao, X., & Xu, G. (2021). Trade or trick? Detecting and characterizing scam tokens on Uniswap decentralized exchange. *Proceedings of the ACM on Measurement and Analysis of Computing Systems*, *5*(3), 1–26. https://doi.org/10.1145/3491051

Zetzsche, D. A., Arner, D. W., & Buckley, R. P. (2020). Decentralized finance. *Journal of Financial Regulation*, *6*(2), 172–203. https://doi.org/10.1093/jfr/fjaa010

Zhou, L., Xiong, X., Ernstberger, J., Chaliasos, S., Wang, Z., Wang, Y., Qin, K., Wattenhofer, R., Song, D., & Gervais, A. (2023). SoK: Decentralized finance (DeFi) attacks. *2023 IEEE Symposium on Security and Privacy (SP)*, 2444–2461. https://doi.org/10.1109/SP46215.2023.10179435